How to treat autism in children in 2026?

Evidence searched on 2026-10-14 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The most supported approaches for treating autism in children in 2026 are **exercise interventions** (reported to improve motor skills) and **multidisciplinary rehabilitation** (for rare genetic syndromes). No direct evidence yet supports music therapy or dance programs for social communication or emotion regulation, though these are under investigation. FDA-approved antipsychotics (e.g., aripiprazole, risperidone) remain available for irritability but are not covered by these findings.

Most important recent finding Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified for overall and gross motor skills. [1]

Immediate context Motor skill interventions are directly supported by recent meta-analysis, while social communication or emotion regulation interventions (e.g., music therapy) are still under investigation. No evidence addresses core ASD symptoms (e.g., social communication deficits) in the retrieved data.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Aripiprazole	"Aripiprazole is indicated for the treatment of: Irritability Associated with Autistic Disorder."	DailyMed
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)."	DailyMed

Latest findings

Art-based interventions and social-emotional outcomes in children with autism spectrum disorder *2026-09-15, Systematic Review and Meta-Analysis* A systematic review and meta-analysis of art-based interventions (e.g., drawing, multimodal programs) in children with ASD found low-certainty evidence of improved social-emotional outcomes, particularly in interventions lasting **9–16 weeks**. Neuroimaging suggested potential mechanisms via frontoparietal attentional control, though direct longitudinal trials are needed to confirm causality [2].

Transcranial direct current stimulation (tDCS) for social function in children and adolescents with ASD *2026 Dec (advance publication), Meta-Analysis* A meta-analysis of **13 RCTs (N=357, ages 4–21 years)** found tDCS significantly improved social functioning (standardized mean difference **−0.51**, p=0.0002) and induced neurofunctional changes, with the largest effects observed via **functional near-infrared spectroscopy (fNIRS)**. Subgroup analyses showed variable efficacy by modality, and heterogeneity was moderate [3].

Accelerated continuous theta burst stimulation (a-cTBS) for ASD *2026-04-29, Multicenter Randomized Controlled Trial* Adverse events were mild and transient [4].

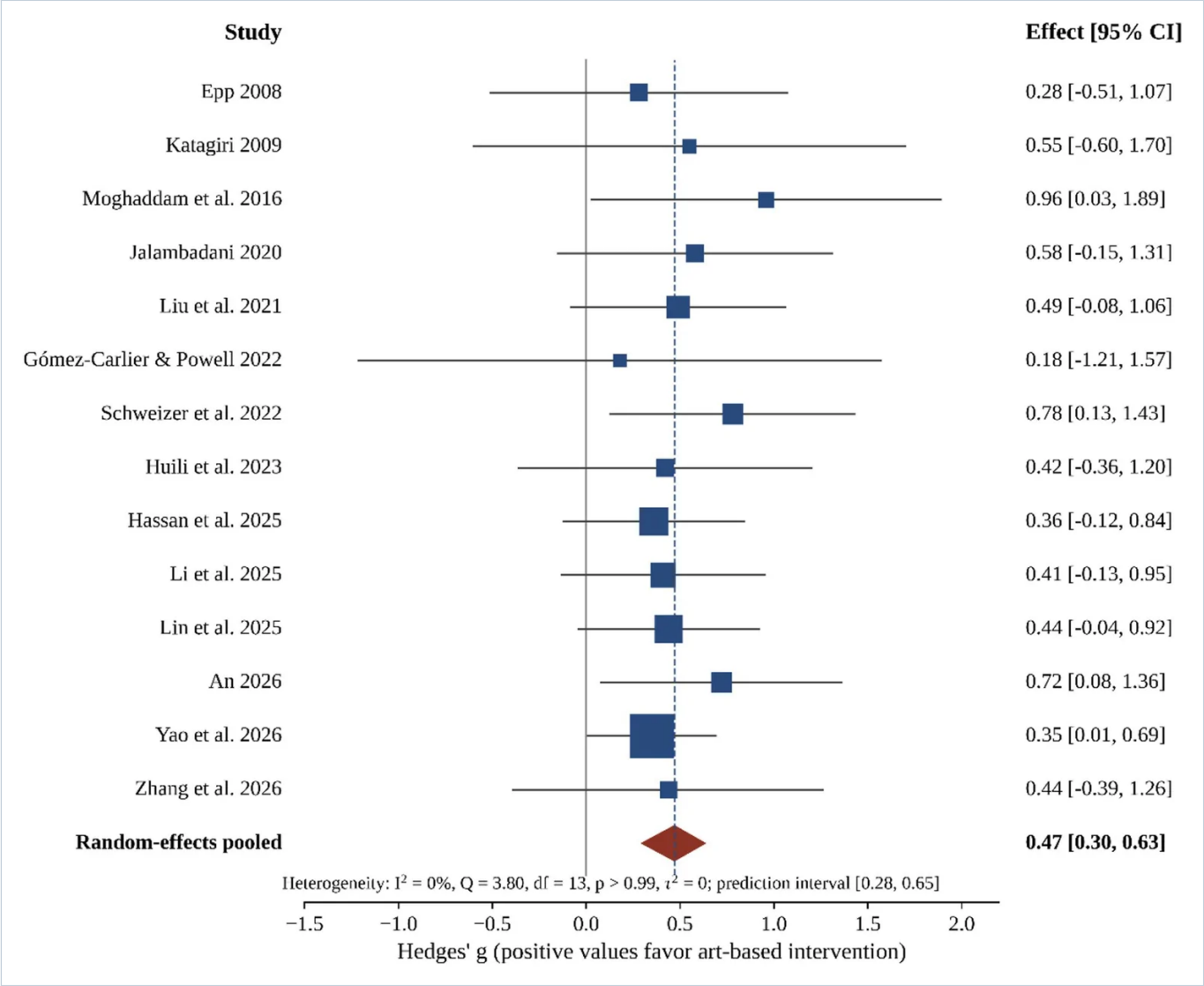
Modified ketogenic diet for ASD *2026-09-11, Randomized Controlled Trial* A **2-month trial** in **young children with ASD** compared a modified ketogenic diet + rehabilitation to rehabilitation alone, finding **greater reductions in ABC (Achenbach Behavior Checklist) and CARS (Childhood Autism Rating Scale) scores** in the intervention group.

Feasibility was acceptable, but long-term effects were not assessed [5].

Extended reality (XR) interventions for ASD *2026 Dec (advance publication), Systematic Review* A review of **115 XR studies** (mostly virtual reality) reported positive outcomes for **engagement, social communication, emotional recognition, and adaptive functioning** in children with ASD. Hybrid systems (e.g., VR + AI) were most common, but heterogeneity in study designs limited preliminary conclusions [6].

Current evidence

Approach	Evidence level	What the evidence shows	Reference
Early Intensive Behavioral Intervention (EIBI)	Evidence-supported but limited	Demonstrated broad, substantial effects on reducing ASD severity in children, with a Number Needed to Treat between 4.1 and 6.9; treatment intensity significantly contributed to outcomes	[9]
Accelerated continuous theta burst stimulation (a-cTBS) targeting the left primary motor cortex	Evidence-supported but limited	Significantly reduced social responsiveness scores (SRS-2) post-intervention and at one-month follow-up in children with ASD (Tan et al., BMJ (Clinical research ed.), 2026 [4]).	[4]
Exercise interventions (e.g., 45-47 min sessions)	Emerging	Significantly improved overall, gross, and fine motor skills in children with ASD; dose-response patterns identified for overall and gross motor skills	[1]
Transcranial direct current stimulation (tDCS)	Emerging	Significantly improved social functioning in children and adolescents with ASD (ages 4-21 years), with distinct neurofunctional modulation effects across measurement modalities	[3]
Modified ketogenic diet combined with rehabilitation training	Emerging	Associated with greater short-term reductions in ASD symptom severity (ABC and CARS scores) compared to rehabilitation alone in young children with ASD	[5]
Art-based interventions (e.g., visual art engagement)	Emerging	Suggested improvements in emotional intelligence-related social-emotional outcomes, though evidence certainty was rated low	[2]
Extended reality (XR) interventions (e.g., VR, AR)	Emerging	Reported positive outcomes related to engagement, social communication, emotional recognition, and adaptive functioning in children with ASD	[6]
Acceptance and Commitment Training (ACT) for parents	Emerging	Improved parental stress, psychological flexibility, parenting competence, and reduced children's emotional/behavioral problems in parents of children with ASD	[10]



What this figure shows: This forest plot compares art-based interventions vs. controls in children aged 3–15 with autism, showing a pooled effect size of 0.47 (95% CI 0.30–0.63) for social-emotional outcomes. The consistent positive effect across studies suggests these interventions may improve emotional intelligence-related skills in autism.

How it helps with treatment options: This figure supports comparing art-based interventions to controls for improving social-emotional outcomes in autistic children, indicating a moderate positive effect across multiple studies.

How to read this figure: Squares show individual study effect sizes (Hedges' g) for art-based interventions; their size reflects study weight. Horizontal lines are 95% confidence intervals; the diamond is the pooled random-effects estimate.

Why this figure was chosen: The model's reason, from the caption: "effects of art-based interventions on EI-related social–emotional outcomes". Checks passed: the paper was judged directly relevant to the question; the article states an open licence; the model's five-part relevance test was passed (helps answer, clinically meaningful, adds to the report, not decorative, shows the kind of information the question asks for); the picture is large enough to read on the page; a vision model confirmed that the image matches its caption and is readable.

Figure 2. Forest plot of the effects of art-based interventions on EI-related social–emotional outcomes in children with autism spectrum disorder. Squares represent study-level effect sizes (Hedges' g), with square size proportional to the inverse-variance weight assigned to each study; horizontal lines denote 95% confidence intervals. The diamond represents the random-effects pooled estimate [SMD = 0.47, 95% CI (0.30, 0.63), p 0.99, τ 2 = 0; 95% prediction interval [0.28, 0.65]. Standard errors were recalculated from the reported group sizes during revision and the figure has been redrawn accordingly.

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Used in practice: a retrieved review or page describes the drug for this condition, with no FDA label or guideline for it (see the category in Key evidence). Doses appear only where an FDA label or source gives them for the use shown; each dose entry is a short summary of the label's dosing section for this condition, with every number checked against the label text (full text in the DailyMed link); individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Key evidence
Aripiprazole	"Aripiprazole is indicated for the treatment of: • Schizophrenia • Irritability Associated with Autistic Disorder • Treatment of Tourette's Disorder" [associated symptom or comorbidity, not the condition itself]	Route: Oral. Start with 2 mg/day in children aged 6 to 17 years. The dose may be increased to 5 mg/day, then further to 10 or 15 mg/day if needed. Adjustments of up to 5 mg/day should be made gradually, with intervals of no less than one week. The usual dose range is 5 to 15 mg/day.	FDA label. DailyMed label FDA-approved for the use shown; label effective 2026-03-05.
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)." [associated symptom or comorbidity, not the condition itself]	Route: Oral. Start with 0.25 mg per day for children weighing less than 20 kg, and 0.5 mg per day for those weighing 20 kg or more. After at least four days, increase to 0.5 mg per day for patients under 20 kg or 1.0 mg per day for those 20 kg or heavier, and maintain this dose for no less than 14 days. Subsequent increases occur every 2 weeks or more, in steps of 0.25 mg per day for patients under 20 kg or 0.5 mg per day for those 20 kg or more. The usual dose range is 0.5 mg to 3 mg per day, and this applies to children and adolescents weighing at least 15 kg.	FDA label: Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years). DailyMed label FDA-approved for the use shown; label effective 2026-06-25.

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Dose / route (if sourced)	Key evidence
Leucovorin	reductions in communication deficits in nonverbal children with ASD (population: nonverbal children with ASD and cerebral folate deficiency) [associated symptom or comorbidity, not the condition itself]	Not stated in the retrieved source for this use	"Certain studies have indicated reductions in communication deficits in nonverbal children with ASD, particularly those with these autoantibodies, following consistent leucovorin use." [7]

Recently approved or updated (FDA, EMA and CDSCO announcements)

No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
Aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
Risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Risperidone tablets are not approved for the treatment of patients with dementia-related psychosis."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: Leucovorin.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07439510	Codabakalner; Placebo	Phase 2/3	Not yet recruiting (2026-10-09)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04903353	Comparison of Risperidone and Aripiprazole	Phase 4	Completed (2026-10-05)	Completed; results not posted/retrieved.
NCT07257939	NTI164; Placebo	Phase 3	Not yet recruiting (2026-09-30)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07285798	KarXT; KarX-EC	Phase 3	Not yet recruiting (2026-09-28)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06229210	Lumateperone	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT03279471	Sertraline; CBT/BIACA	Phase 2	Completed (2026-09-23)	Completed; results posted (outcome not retrieved here).
NCT07424404	KarXT; KarX-EC	Phase 3	Recruiting (2026-09-22)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04630847	Fecal Microbiota Transplant	Phase 1	Not yet recruiting (2026-09-18)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07439042	buspirone; Placebo	Phase 4	Recruiting (2026-09-17)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT03514784	BB-12 with LGG (Higher Dose); Placebo	Phase 1/2	Completed (2026-09-14)	Completed; results posted (outcome not retrieved here).
NCT05626959	NTI164	Phase 2/3	Completed (2026-09-10)	Completed; results not posted/retrieved.
NCT06368726	tDCS or Transcranial Direct Current Stimulation; Risperidone	Phase 1/2	Completed (2026-09-10)	Completed; results not posted/retrieved.

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Wang et al. (2026)**, meta-analysis: “Exercise significantly improved overall, gross, and fine motor skills in children with ASD.” [1]
2. **He et al. (2026)**, systematic review: “Exploratory subgroup analyses showed no material difference between multimodal and drawing-focused programs (p = 0.79) and a larger point estimate for interventions lasting 9-16 weeks, but these moderator findings were descriptive and should be interpreted cautiously.” [2]
3. **Hao et al. (2026)**, systematic review: “There was a significant increase in XR autism research after 2020, with virtual reality (VR) as the largest modality (n = 61, 53.0%), followed by augmented reality (AR) (n = 36, 31.3%).” [6]
4. **Zhao et al. (2026)**, meta-analysis: “Thirteen RCTs were included (N = 357, ages 4-21 years). tDCS significantly improved social functioning (SMD = -0.51, I2 = 0%, p = 0.0002) and induced significant overall neurofunctional modulations (SMD = 1.03, I2 = 71.9%, p = 0.004).” [3]
5. **Chan et al. (2026)**, meta-analysis: “No significant difference in ASD risk was found between high-dose and low-dose exposure, and paternal antidepressant use around conception was not linked to ASD.” [11]

Conflicting evidence

No direct disagreement was retrieved between studies on the efficacy of established or emerging treatments for autism in children. However, **Sirao et al. (2026)** found behavioral strategies and exercise to be safe, low-cost, and effective for sleep improvement, while **Liu et al. (2026)** reported short-term feasibility of a modified ketogenic diet but did not compare it directly to behavioral interventions for sleep outcomes. No head-to-head trials were identified.

What remains under investigation

- **Efficacy of extended reality (XR) interventions**: While positive outcomes were reported for engagement, social communication, and adaptive functioning, no randomized controlled trials were retrieved to establish direct efficacy in children with ASD [6].
- **Long-term effects of transcranial direct current stimulation (tDCS)**: Although tDCS improved social functioning in short-term studies, long-term sustainability and optimal parameters (e.g., electrode placement, current intensity) remain unclear [3].
- **Feasibility of deep brain stimulation (DBS) in routine care**: While **Shi et al. (2026)** reported sustained core symptom reduction in a single case over 12 months, broader applicability, safety in diverse populations, and cost-effectiveness are unproven.
- **Effectiveness of occupational therapy-based water competency interventions**: The **AquOTic trial [8]** aims to assess implementation determinants and costs but has not yet established efficacy for water safety or motor skills in ASD.
- **Association between sleep spindle characteristics and ASD severity**: While **Erçelebi et al. (2026)** identified reduced spindle activity as a potential neurophysiological marker, causal mechanisms and therapeutic targeting remain speculative.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- 5 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.

- The web search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.

Sources

1. Wang T, Dong X, Jia Y, et al. Dose-response relationship between exercise interventions and motor skills in children with autism spectrum disorder: a systematic review and multilevel meta-analysis. *European journal of pediatrics*. 2026-09-18. PMID 42758334. DOI 10.1007/s00431-026-07421-4.
2. He L, He R, Liu G, et al. Art-based interventions and social-emotional outcomes in children with autism spectrum disorder: a systematic review and meta-analysis with a proposed neurodevelopmental framework. *Frontiers in human neuroscience*. 2026-09-15. PMID 42812367. DOI 10.3389/fnhum.2026.1941559.
3. Zhao H, Niu H, Jiang Y, et al. Efficacy and neural mechanisms of transcranial direct current stimulation for social function in children and adolescents with autism spectrum disorder: A systematic review and meta-analysis. *Journal of psychiatric research*. 2026 Dec. PMID 42759400. DOI 10.1016/j.jpsychires.2026.09.019.
4. Tan H, Ren T, Cao A, et al. Accelerated continuous theta burst stimulation targeting left primary motor cortex for children with autism spectrum disorder: multicentre randomised sham controlled trial. *BMJ (Clinical research ed.)*. 2026 Apr 29. PMID 42055586. DOI 10.1136/bmj-2025-086295.
5. Liu L, Zhao Q, Zhou J, et al. Evaluating the therapeutic efficacy of a modified ketogenic diet in children with autism spectrum disorder: a randomized controlled trial. *BMC pediatrics*. 2026-09-11. PMID 42763398. DOI 10.1186/s12887-026-07708-3.
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7. Howard C, Mekhail J, Ravikoff LM, et al. Clinical concerns and considerations for leucovorin use in autism spectrum disorder. *Current opinion in pediatrics*. 2026 Apr 1. PMID 41670435. DOI 10.1097/MOP.0000000000001547.
8. Kemp E, Juckett L, Sansone IO, et al. AquOTic: protocol for a type 1 hybrid effectiveness-implementation trial of an occupational therapy-based water competency intervention for children on the autism spectrum. *BMJ open*. 2026-08-04. PMID 42552003. DOI 10.1136/bmjopen-2026-117908.
9. Eldevik S, Strømgren B, Eikeseth S, et al. Clinically Significant Outcomes of Early Intensive Behavioral Intervention for Children With Autism Spectrum Disorders: An Individual Participant Data Meta-Analysis. *Autism research : official journal of the International Society for Autism Research*. 2026 Jan. PMID 41502379. DOI 10.1002/aur.70169.
10. Li SN, Chien WT. Acceptance and Commitment Training for Parents of Children With Autism Spectrum Disorder: A Randomized Clinical Trial. *JAMA network open*. 2026 Jan 2. PMID 41505131. DOI 10.1001/jamanetworkopen.2025.52693.
11. Chan JKN, Zhong AHF, Lam JYH, et al. Maternal and paternal antidepressant use before and during pregnancy and offspring risk of neurodevelopmental disorders: a systematic review and meta-analysis. *The lancet. Psychiatry*. 2026 Jun. PMID 42134364. DOI 10.1016/S2215-0366(26)00089-1.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- FDA drug label: Aripiprazole (label effective 2026-03-05)
- FDA drug label: Risperidone (label effective 2026-06-25)

Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.

Figure credits

Figure 2: Art-based interventions and social-emotional outcomes in children with autism spectrum disorder: a systematic review and meta-analysis with a proposed neurodevelopmental framework. Systematic Review, systematic-review. PMID 42812367, DOI 10.3389/fnhum.2026.1941559, PubMed Central (PMC13619913). Cited in this report as [2]. Licence: CC BY 4.0; original image, unaltered.

Each figure shows one study's result; it does not by itself establish that a treatment works or justify a recommendation.